

RadioScreen

An Educational AI Prototype for Chest X-ray Screening

■ **EDUCATIONAL / RESEARCH PROTOTYPE ONLY — NOT a certified medical device, NOT for real diagnosis, and NOT a substitute for a licensed physician or radiologist. All data used in this prototype is synthetically generated.**

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1. Problem Statement

Chest X-rays are among the most widely used diagnostic imaging tools worldwide, but radiologist availability is limited in many regions, and reading volumes are high. AI-assisted screening tools have been studied extensively in the research literature (e.g., the CheXNet study on the NIH ChestX-ray14 dataset) as a way to help flag images that may warrant closer review — always as a support tool for clinicians, never as a replacement for one.

2. Project Scope and Ethical Framing

RadioScreen is an educational, portfolio-oriented exploration of this idea. It is explicitly scoped as a research prototype: it uses synthetically generated images rather than real patient data (real datasets such as NIH ChestX-ray14, CheXpert, or the RSNA Pneumonia dataset require signed data-use agreements), and every interface and API response carries a persistent disclaimer. The goal is to demonstrate the engineering pipeline — data handling, feature extraction or deep learning, an API, and a review interface — end to end, not to produce a clinically usable diagnostic system.

3. System Architecture

The project is implemented in three layers, mirroring how a real clinical decision-support tool would be structured, while keeping the scope honest about what is demonstrated versus what would require a real dataset and clinical validation.

3.1 Proof of Concept (runnable, synthetic data)

A lightweight, dependency-light pipeline: synthetic grayscale images are generated with localized brighter regions standing in for opacity patterns, hand-crafted features are extracted (intensity statistics and edge density), and a Random Forest classifier is trained and evaluated. This runs end to end with only Scikit-Learn and NumPy, and its output is included below.

3.2 Reference Deep Learning Design (production-oriented)

A separate reference script documents how the system would be built for a real deployment: transfer learning on a DenseNet-121 backbone pretrained on ImageNet — the same backbone architecture used in the CheXNet study — fine-tuned on a licensed public chest X-ray dataset. This script is provided as a design reference and was not executed in this environment, since it requires a GPU deep-learning stack and a licensed dataset not available in the development sandbox.

3.3 Backend API

A FastAPI service exposing a single POST /screen endpoint: it accepts an uploaded image, extracts the same features used in the proof of concept, and returns an illustrative 'abnormality likelihood' score together with a mandatory disclaimer field in every response.

3.4 Frontend — Review Interface

A single-page clinical review interface: an image viewer, an upload dropzone, and a result panel showing a likelihood gauge. A non-dismissible disclaimer banner is always visible at the top of the screen. If the backend is unreachable, the interface clearly labels its output as a locally simulated score rather than silently failing.

4. Proof of Concept — Code and Output

4.1 Execution Output

Note on these metrics: the near-perfect scores reflect that the synthetic 'abnormal' images were generated with a clearly separable injected pattern — this validates that the pipeline works correctly end to end, but these numbers say nothing about real-world diagnostic performance, which would require evaluation on a real, clinically annotated dataset.

5. Repository Structure

■■■ index.html Clinical review interface (HTML/CSS/JS)

6. Next Steps Toward a Real Prototype

- Obtain a signed data-use agreement for a public dataset (NIH ChestX-ray14, CheXpert, or RSNA Pneumonia) and replace the synthetic data generator.
- Train the DenseNet-121 reference model on real, licensed data with proper train/validation/test splits by patient (not just by image) to avoid data leakage.
- Evaluate with clinically meaningful metrics (sensitivity at a fixed specificity, calibration curves) rather than raw accuracy.
- Any path toward real-world use would require IRB-style ethical review, clinical validation with radiologists, and regulatory clearance — this project stops deliberately short of that scope and is presented as a learning artifact.

■ **Final reminder: RadioScreen is an educational and research prototype only. It has not been clinically validated, is not regulatory-approved, and must never be used for real diagnostic decisions.**